Alan Chen

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Experience

Scrawlr Sep 2025 - Dec 2025
Software Engineer, Platform & Infrastructure Vancouver, BC

- Engineered and deployed a production agentic system using an MCP server and tools, codebase context, and Puppeteer to diagnose incidents, generate corrective PRs, and validate fixes, reducing **MTTR by over 80%**.
- Instrumented the system's outputs, tool calls, and intermediate steps; enforced browser domain allowlists and scoped repository permissions, requiring engineer-approved PRs for production changes, with fallback and escalation paths.
- Owned AWS infrastructure with Terraform and CI/CD with GitHub Actions/CircleCI; built isolated per-branch QA environments on shared infrastructure, removing a testing bottleneck and increasing **QA throughput by 50%**.
- Led end-to-end delivery of Lodgr's property management platform across frontend, backend, and QA under a convention deadline; resolved OAuth redirect issues across subdomains to enable host-guest messaging.

UKG May 2025 - Aug 2025
Software Engineer, Mobile Platform Toronto, ON

- Implemented proxy routing with service-discovery fallback for critical, high-traffic services, replacing repeated discovery requests and reducing service-registry load.
- Built mobile observability with Splunk dashboards, distributed tracing, and PagerDuty/Slack alerts, surfacing client side failures and accelerating incident response.
- Redesigned UKG Pro's mobile release pipelines to build and test rollback artifacts for App Store and Play Store submission, meeting forward-only versioning requirement.
- Refactored UKG Pro's local database modules and Ionic-to-native bridges across Ionic, Kotlin, and Swift, improving maintainability of the mobile platform.

Metergy Solutions Jan 2024 - May 2025
Software Engineer Markham, ON

- Owned 0-to-1 delivery of an offline-first software platform, translating installation-team requirements into architecture, local persistence, and authenticated cloud sync; improved efficiency by an **estimated 50%**.
- Built a backend ingestion service for device-initiated HTTP uploads, replacing FTP polling and cron jobs; raised reliability from **70% to 99.9%** across **10,000+ IoT meters**.
- Built an AI OCR pipeline for utility billing that extracted bill data and routed low-confidence results for human review, reducing manual processing.
- Modernized business-critical Access tools into a modular Django API and Vue application for concurrent team access, improving reliability and **halving runtime** for jobs such as bulk email.

Education

University of Waterloo 2021 - 2026
Honours Bachelor of Computer Science, Co-op

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, C#, Go, C++, Kotlin, Swift
Backend and Data: FastAPI, Django, Node.js (Fastify), Spring Boot, PostgreSQL, MySQL, Redis
Cloud and CI/CD: AWS, GCP, Docker, Kubernetes, Terraform, Pulumi, GitHub Actions, CircleCI
AI and Automation: Agentic workflows, Model Context Protocol (MCP), agent skills, tool orchestration
Observability: Sentry, Splunk, Prometheus, Grafana
Frontend and Mobile: React, Vue.js, Ionic Framework, SwiftUI, Jetpack Compose